

Quiz 4

$$\alpha_n = \frac{(2n-1)\pi}{2a}$$

$$\beta_m = \frac{(2m-1)\pi}{2b}$$

$$V_z(x, y) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \cos(\alpha_n x) \cos(\beta_m y)$$

$$\frac{\partial V_z}{\partial x} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (-\alpha_n C_{nm} \sin(\alpha_n x) \cos(\beta_m y))$$

$$\frac{\partial^2 V_z}{\partial x^2} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} -\alpha_n^2 C_{nm} \cos(\alpha_n x) \cos(\beta_m y)$$

$$\frac{\partial V_z}{\partial y} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (-\beta_m C_{nm} \cos(\alpha_n x) \sin(\beta_m y))$$

$$\frac{\partial^2 V_z}{\partial y^2} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} (-\beta_m^2 C_{nm} \cos(\alpha_n x) \cos(\beta_m y))$$

$$\frac{\partial^2 V_z}{\partial x^2} + \frac{\partial^2 V_z}{\partial y^2} = \frac{\Delta p}{\mu L}$$

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} -\alpha_n^2 C_{nm} \cos(\alpha_n x) \cos(\beta_m y) + \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} -\beta_m^2 C_{nm} \cos(\alpha_n x) \cos(\beta_m y) = \frac{\Delta p}{\mu L}$$

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} -(\alpha_n^2 + \beta_m^2) C_{nm} \cos(\alpha_n x) \cos(\beta_m y) = \frac{\Delta p}{\mu L}$$

$$-(\alpha_n^2 + \beta_m^2) C_{nm} = \beta_m$$

$$C_{nm} = -\frac{\beta_m}{\alpha_n^2 + \beta_m^2}$$

$$\int_{-a}^a \int_{-b}^b \frac{\Delta P}{uL} \cos(\alpha p x) \cos(\beta q y) dy dx =$$

$$\int_{-a}^a \int_{-b}^b \left(\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} B_{nm} \cos(\alpha n x) \cos(\beta m y) \right) \cos(\alpha p x) \cos(\beta q y) dy dx$$

$$\int_{-a}^a \int_{-b}^b \frac{\Delta P}{uL} \cos(\alpha p x) \cos(\beta q y) dy dx = B_{pq} \left(\int_{-a}^a \cos^2(\alpha p x) dx \right) \left(\int_{-b}^b \cos^2(\beta q y) dy \right)$$

$$\int_{-a}^a \cos^2(\alpha p x) dx = \int_{-a}^a \frac{1 + \cos(2\alpha p x)}{2} dx = \frac{1}{2} \int_{-a}^a 1 dx + \frac{1}{2} \int_{-a}^a \cos(2\alpha p x) dx$$

$$= a$$

$$\int_{-b}^b \cos^2(\beta q y) dy = b$$

$$\frac{\Delta P}{uL} \left(\int_{-a}^a \cos(\alpha p x) dx \right) \left(\int_{-b}^b \cos(\beta q y) dy \right)$$

$$\int_{-a}^a \cos(\alpha p x) dx = \frac{\sin(\alpha p x)}{\alpha p} \Big|_{-a}^a = \frac{2 \sin(\alpha p a)}{\alpha p}$$

$$\sin(\alpha p a) = \sin\left(\frac{(2p-1)\pi}{2}\right) = (-1)^{p-1} = \frac{2(-1)^{p-1}}{\alpha p}$$

$$\int_{-b}^b \cos(\beta q y) dy = \frac{2(-1)^{q-1}}{\beta q}$$

$$\frac{\Delta P}{uL} \left(\frac{2(-1)^{p-1}}{\alpha p} \right) \left(\frac{2(-1)^{q-1}}{\beta q} \right) = \frac{\Delta P}{uL} \frac{4(-1)^{p+q}}{\alpha p \beta q}$$

$$B_{pq} = \frac{\Delta P}{uL} \frac{4(-1)^{p+q}}{\alpha \beta \alpha p \beta q} = \frac{(2p-1)(2q-1)\pi^2}{4ab}$$

$$= \frac{\Delta P}{uL} \frac{16(-1)^{p+q}}{(2p-1)(2q-1)\pi^2}$$

$$B_{nm} = \frac{16 \Delta P}{\mu L \pi^2} \frac{(-1)^{n+m}}{(2n-1)(2m-1)}$$

$$C_{nm} = \frac{-16 \Delta P}{\mu L \pi^2} \frac{(-1)^{n+m}}{(2n-1)(2m-1)} \frac{1}{a^2 + b^2}$$

$$a^2 + b^2 = \frac{(2n-1)^2 \pi^2}{4a^2} + \frac{(2m-1)^2 \pi^2}{4b^2}$$

$$= \frac{\pi^2}{4} \left(\frac{(2n-1)^2}{a^2} + \frac{(2m-1)^2}{b^2} \right)$$

$$C_{nm} = \frac{-64 \Delta P}{\mu L \pi^4} \frac{(-1)^{n+m}}{(2n-1)(2m-1) \left(\frac{(2n-1)^2}{a^2} + \frac{(2m-1)^2}{b^2} \right)}$$

$$v_z = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \left(\frac{-64 \Delta P}{\mu L \pi^4} \frac{(-1)^{n+m}}{(2n-1)(2m-1) \left(\frac{(2n-1)^2}{a^2} + \frac{(2m-1)^2}{b^2} \right)} \right) \cos\left(\frac{2n-1}{2a} \pi x\right) \cos\left(\frac{2m-1}{2b} \pi y\right)$$

$$\frac{a}{b} = 1 \quad a = b$$

$$\frac{v_z}{\left(\frac{-\Delta P a^2}{\mu L \pi^4} \right)} = 64 \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^{n+m}}{(2n-1)(2m-1) \left((2n-1)^2 + (2m-1)^2 \right)} \cos\left(\frac{2n-1}{2a} \pi x\right) \cos\left(\frac{2m-1}{2a} \pi y\right)$$

$$= 64 \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^{n+m}}{(2n-1)(2m-1) \left((2n-1)^2 + (2m-1)^2 \right)} \cos\left(\frac{2n-1}{2} \pi \frac{x}{a}\right) \cos\left(\frac{2m-1}{2} \pi \frac{y}{a}\right)$$

